



U.S. General Services
Administration

Chapter 6.2A: Vector Database Selection

US General Service Administration
AI Community of Practice - March 2026

Choosing by Scale

- Under 100K vectors: Chroma or pgvector for simplicity
 - Over 1M vectors: evaluate managed vs self-hosted options
 - Managed services trade cost for operational simplicity
 - Self-hosted options offer control and lower marginal cost
 - Match database choice to team maturity and scale needs
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The Big Four -- Milvus, Weaviate, Pinecone, Qdrant

Milvus



billions of vectors

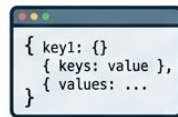


GPU acceleration



distributed clusters

Weaviate



GraphQL interface



built-in vectorization



hybrid search

Pinecone



fully managed



zero infrastructure



automatic scaling

Qdrant



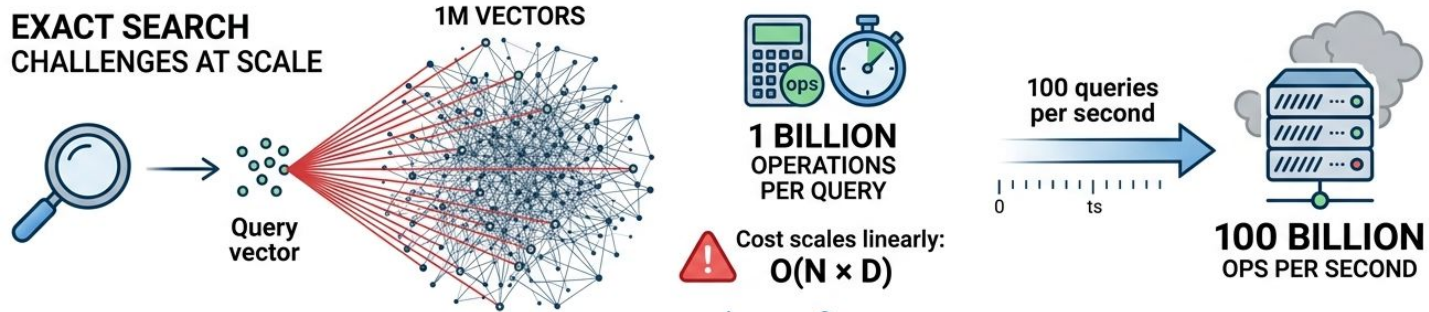
Rust-based



exceptional filtered
search performance

Why Exact Search Breaks at Scale

EXACT SEARCH CHALLENGES AT SCALE



EXACT MATCH  guaranteed perfection

1M VECTORS

EXACT
(Brute Force)

100% Accuracy, Slow

APPROXIMATE
(ANN Algorithms)

~95% Accuracy, Fast

BRUTE FORCE COMPARES AGAINST EVERY VECTOR

APPROXIMATE SEARCH SOLUTIONS

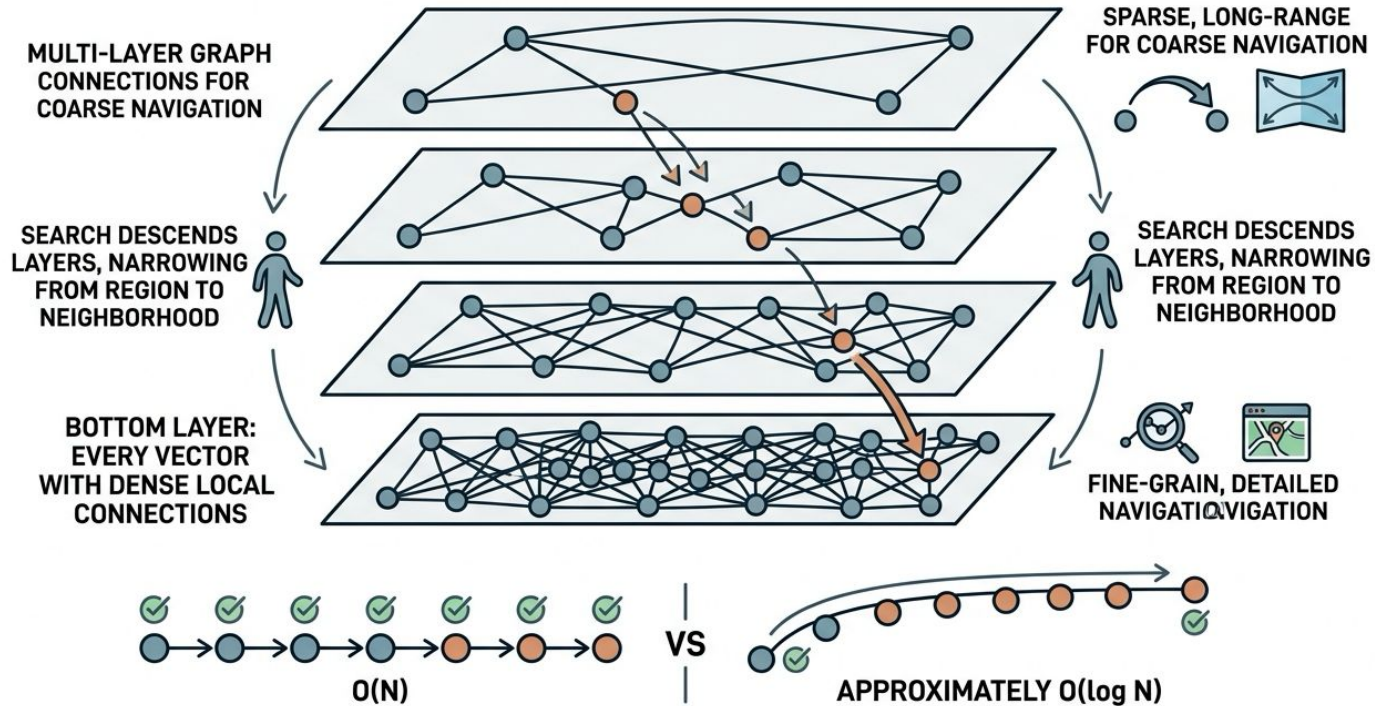



SLIGHT
ACCURACY LOSS


MASSIVE
SPEEDUP

TRADE SMALL ACCURACY LOSS FOR MASSIVE SPEED

HNSW Algorithm

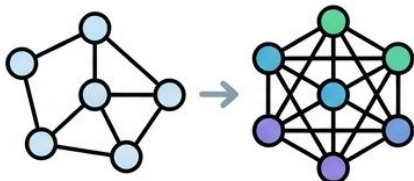


Tuning HNSW vector search



M (CONNECTIVITY)

LINKS PER NODE: $M=16$
(STRONG DEFAULT)



Low M

High M



HIGHER M:
MORE MEMORY,
SLOWER QUERIES



BETTER
RECALL



**efCONSTRUCTION
(BUILD QUALITY)**

INVESTMENT

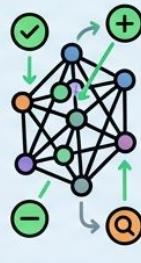


ONE-TIME
COST



INDEX
BUILD

BENEFIT
FOREVER

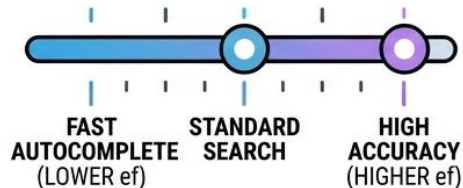


HIGH-QUALITY
CONNECTIONS
CONNECTIONS

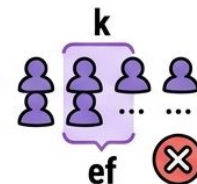
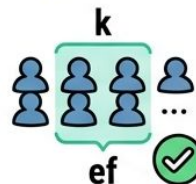


**ef (RUNTIME
SEARCH DEPTH)**

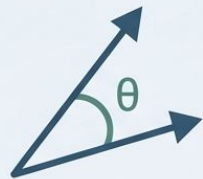
ADJUSTABLE PER QUERY



**CRITICAL RULE: $ef \geq k$
(RESULTS REQUESTED)**



Distance Metrics -- Cosine, L2, and Dot Product



Cosine Similarity



MEASURES ANGLE

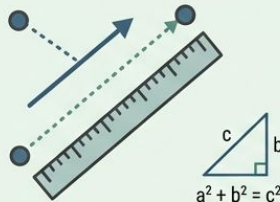


IGNORES MAGNITUDE



MOST TEXT EMBEDDING MODELS OPTIMIZE FOR COSINE SIMILARITY

BEST FOR TEXT: Focuses on direction, ignoring training noise in vector length.



L2 (Euclidean) Distance



MEASURES SPATIAL SEPARATION



INCLUDES MAGNITUDE



USEFUL WHEN MAGNITUDE MATTERS



(e.g., Image Embeddings, custom embeddings with semantic length)



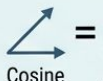
Dot Product



FASTEST COMPUTATION



SKIPS NORMALIZATION CALCULATION



Cosine



Dot Product

IDENTICAL TO COSINE SIMILARITY IF VECTORS ARE NORMALIZED



PURE SPEED WIN FOR PRE-NORMALIZED EMBEDDINGS

GUIDELINES

IF UNSURE



RUN BENCHMARK

IF PRE-NORMALIZED VECTORS



USE DOT PRODUCT

IF OPTIMIZED FOR COSINE (e.g., specific models)



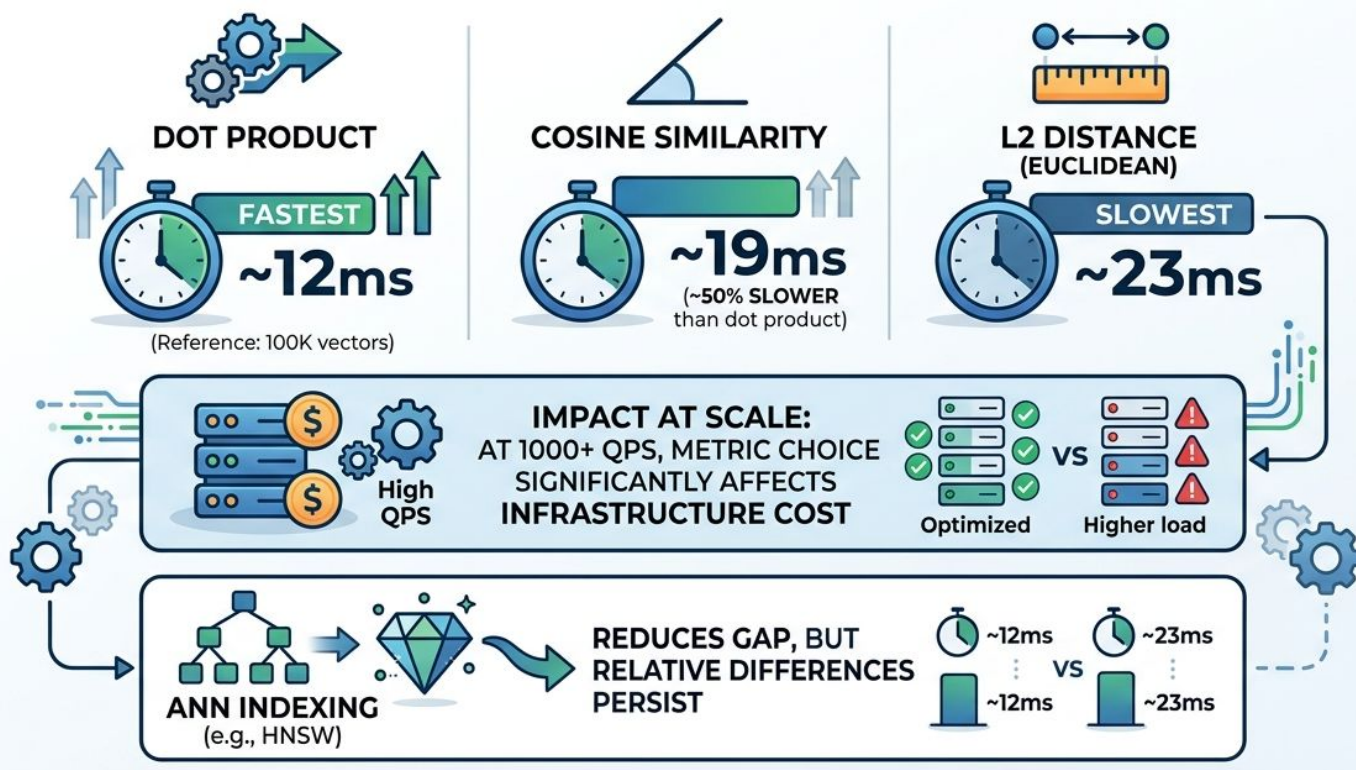
USE COSINE SIMILARITY

IF MAGNITUDE MATTERS



USE L2 DISTANCE

Benchmarking Metrics -- The Numbers That Matter



The Practical Decision Framework

STARTING POINT



M = 16



EFCONSTRUCTION = 200



EF = 100



DOT PRODUCT
METRIC

FOR NORMALIZED EMBEDDINGS

ADJUST M FOR RECALL



RECALL < 90%



BUMP M
TO 24



THEN
32



AVOID > 32
(HIGH MEMORY COST)

DIMINISHING RETURNS

INVEST IN EFCONSTRUCTION



ONE-TIME COST



QUERY BENEFITS
FOREVER



BATCH
INGESTION



BATCH
INCREASE

BE GENEROUS

DYNAMIC EF TUNING

search... | 🔍

AUTOCOMPLETE
(LOW LATENCY)

LOWER
EF



important search... | 🔍

IMPORTANT SEARCH
(HIGH ACCURACY)

HIGHER
EF



TUNE PER QUERY AT RUNTIME

VALIDATE WITH REAL QUERIES

Search... | 🔍
200-500 REAL QUERIES



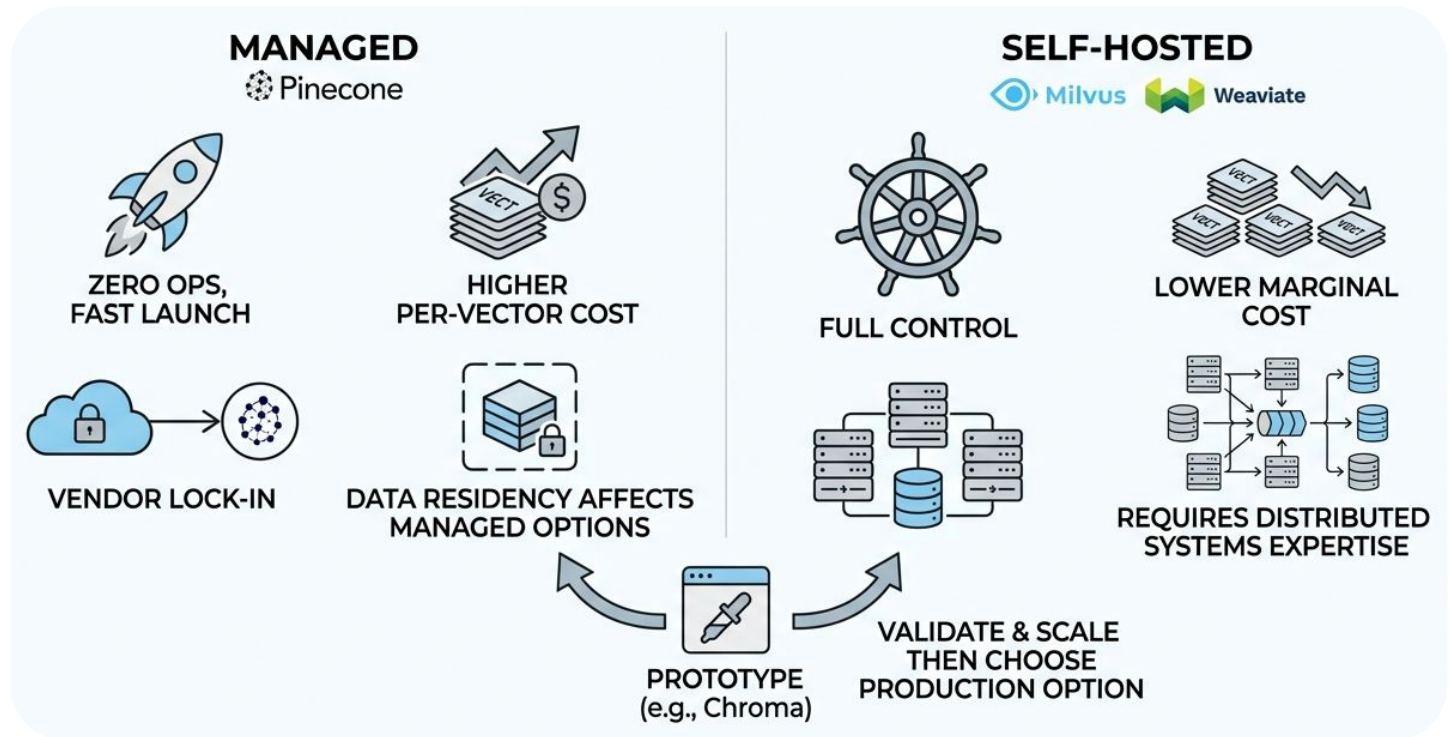
MEASURE
RECALL@10
MEASURE
LATENCY



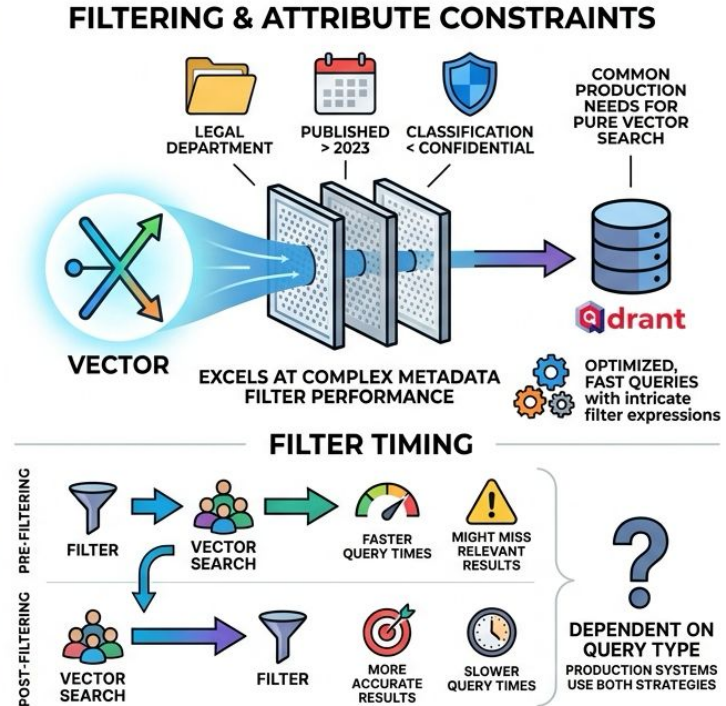
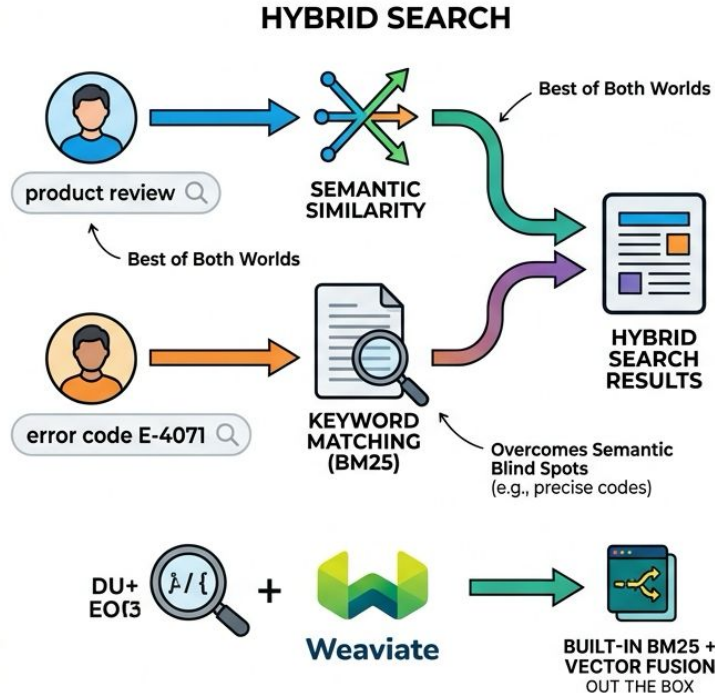
LET THE NUMBERS GUIDE TUNING

VALIDATE WITH YOUR ACTUAL DATA

Managed vs Self-Hosted -- The Real Tradeoff



Filtering and Hybrid Search Capabilities



From Selection to Production Deployment

- Selection criteria are necessary but not sufficient for production
 - Deployment involves replication, monitoring, and failure recovery
 - Index management and data pipeline integration require planning
 - Chapter 6.2B covers production vector database deployment
 - Key takeaway: match database to workload, then tune with real data
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